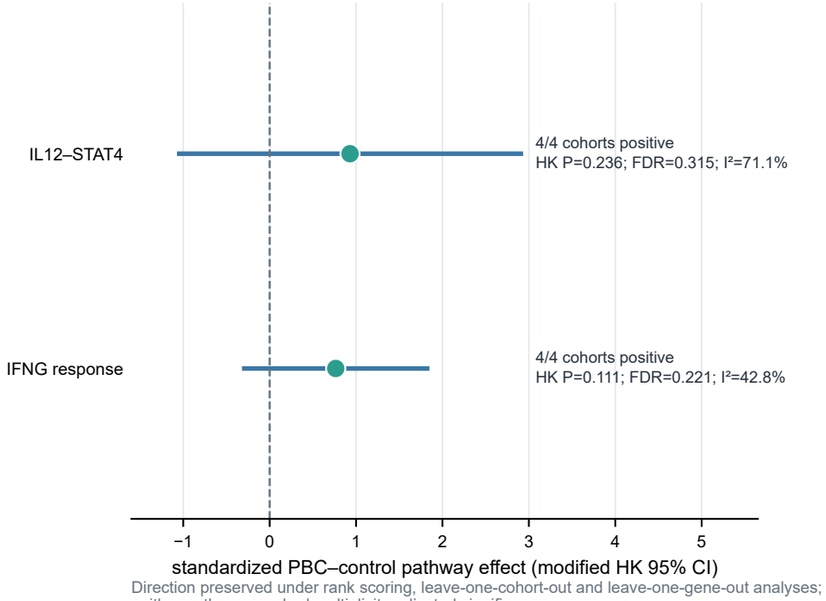
# Orthogonal molecular evidence and a guarded translational path for the IL12RB2–TYK2 axis

Directional multi-omics support is separated from clinical efficacy, safety and druggability claims

## A Bulk transcriptomic pathway meta-analysis

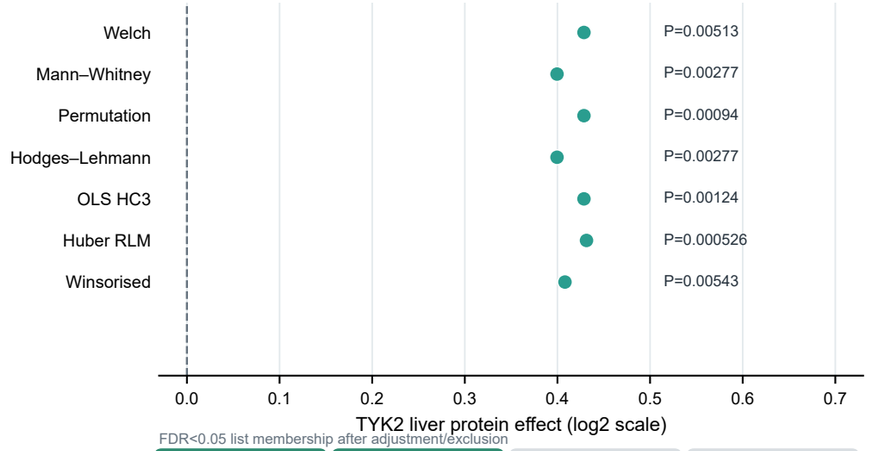
Four independent cohorts; prespecified robust-z score




Direction preserved under rank scoring, leave-one-cohort-out and leave-one-gene-out analyses; neither pathway reached multiplicity-adjusted significance.

## B PXD083842 proteomic robustness and confounding

PBC n=14 versus healthy n=9




FDR<0.05 list membership after adjustment/exclusion

| | | | |
|---|---|---|---|
| **Unadjusted** | **Cirrhosis** | Age | Sex |
| BMI | Age+sex | No MASLD | vs MASLD |

**Critical: age has minimal overlap; recruitment stream is exactly confounded with disease. Adjusted non-significant coefficients were not released. Supportive, not disease-specific replication.**

## C Drug and clinical evidence matrix

Open Targets / ChEMBL / ClinicalTrials.gov / labels

| | Target / mechanism | Clinical-stage modality | PBC trial evidence | Label / safety evidence | Genetic-direction match |
|---|---|---|---|---|---|
| IL12RB2 direct | + | — | — | — | + |
| TYK2 / deucravacitinib | + | + | — | ! | + |
| IL12B / ustekinumab | + | + | — | ! | ! |
| JAK1/2 / baricitinib | + | + | — | — | ! |

+ available/supportive ! qualified risk — absent or negative

**PBC counterevidence: ustekinumab 0/20 achieved week-12 ALP response; baricitinib trial terminated after n=2. No TYK2-directed PBC trial found.**

## D Safety and druggability gates

Green does not denote efficacy; it denotes passage of that specific gate

### TYK2 / deucravacitinib

| | |
|---|---|
| Mechanistic and genetic direction | Matched |
| Selective approved TYK2 modality | Available |
| Direct efficacy evidence in PBC | Absent |
| ALT / AST and DILI monitoring | Required |
| Serious infection and latent TB | Core gate |
| Child–Pugh C | Not recommended |

**CONDITIONAL GO**

Preclinical / early proof-of-mechanism only

### Direct IL12RB2 small-molecule route

| | |
|---|---|
| Experimental receptor structures | 8XRP / 8YI7 |
| Ligand-bound small-molecule pocket | Absent |
| ChEMBL IL12RB2 activity record | Absent |
| Validated docking benchmark | Absent |
| Receptor-selective biologic concept | Research only |

**NO-GO for docking**

Reassess if a validated pocket or ligand emerges

## E Final evidence grade

Claim strength is capped by the weakest required evidence layer

**1 Genetic target support** — MODERATE–STRONG

IL12RB2 locus/eQTL convergence; TYK2 human loss-of-function direction

**2 Translational hypothesis** — CONDITIONAL

Bulk direction is consistent but non-significant; proteomic signal is confounded; selective TYK2 is testable

**3 Clinical recommendation** — NOT SUPPORTED

No direct TYK2 PBC trial; prior pathway trials negative/inconclusive; PBC-specific efficacy and safety remain unknown

**Final claim: genetically supported target axis with a guarded experimental drug hypothesis**

All estimates and decisions are read from locked project tables. Database presence denotes evidence availability, not therapeutic efficacy. Audits current to 14–16 October 2026.